Exploratory generalized-Bayes logistic-regression baseline

The sweep of sec. 10.5 characterizes the heuristic server axis. This exploratory baseline instead probes the client-side axis in a synthetic point-estimate logistic-regression setting.

Exploratory... (part 2)

The cited robust-Bayes and federated- learning results provide the source context (McMahan et al. 2017; Ashman et al. 2022; Bui et al. 2018);

this finite experiment does not reproduce their source protocol or inherit their conclusions unconditionally.

Exploratory... (part 3)

An exploratory federated logistic-regression point-estimate proxy is trained under client label contamination. Standard clients use the NLL gradient with L2 coefficient 0.05; the comparison uses the RCCE gradient at $q = 1.00$ with L2 coefficient 0.10.

Exploratory... (part 4)

The legacy `divergence="KLD"` and `divergence="AR"` arguments select those coefficients for compatibility; this module does not evaluate KL or Alpha-Rényi divergence between weight distributions, represent posterior covariance, or implement a FedGVI generalized-posterior objective.

Exploratory... (part 5)

The estimand is therefore a composite contrast between two joint loss-and-shrinkage configurations, not an isolated RCCE effect. RCCE's $q \to 0$ gradient limit recovers the NLL gradient, while the separate categorical recovery identities remain in sec. 10.1.

Exploratory… (part 6)

The corresponding bounded-influence claim remains source-conditional on FedGVI's assumptions (Mildner et al. 2025); this finite curve does not test that theorem. The losses connect the density-power line (Basu et al. 1998), generalized cross-entropy (Zhang and Sabuncu 2018),

Exploratory... (part 7)

and the generalized-Bayes objective (Bissiri et al. 2016; Knoblauch et al. 2022).

Exploratory... (part 8)

The displayed configuration uses $q = 1.00$ and 200 points per class per client. Those are configured inputs, not parameters selected by this report. Only the peak-margin contamination level is selected within the displayed contamination sweep, by the maximum composite-configuration accuracy difference.

Exploratory... (part 9)

The neighboring-q sensitivity check is descriptive stability evidence, not leakage-free calibration or confirmatory model selection.

Exploratory... (part 10)

As contamination rises, the two configuration curves remain close at lower rates and separate over part of the moderate-to-high range. The largest displayed composite-configuration mean difference occurs at 0.35 contamination, with margin 0.028.

Exploratory... (part 11)

Neighboring tested loss parameters retain a thresholded separation at more than one contamination level, and the plotted bands are seed-level 95% bootstrap intervals.

Untagged PDF derivative | HTML reader

Exploratory... (part 12)

At the highest swept rate, 0.40, both configurations decline and no reliable ordering remains. Retaining that endpoint is necessary for the finite-sweep interpretation. Together with the within-sweep peak-contamination selection,

Exploratory... (part 13)

it makes this an exploratory conditional pattern rather than a precalibrated or confirmatory robustness result.

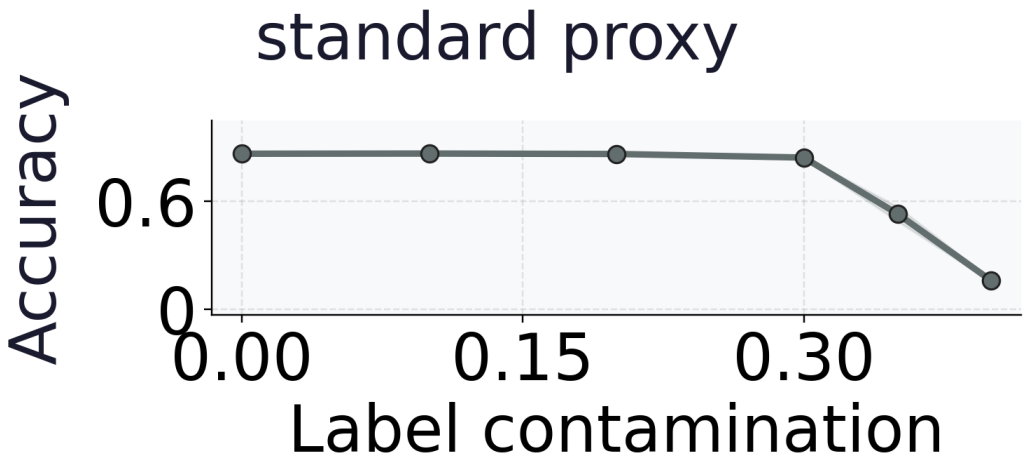
Exploratory... (part 14)

The separation in this small logistic-regression setting is modest. The recovery identities establish implementation compatibility at the named limit (sec. 10.1); the bounded-influence result comes from the cited FedGVI theorem under its assumptions (Mildner et al. 2025),

Exploratory... (part 15)

not from this curve's gap. Source-dataset parity, leakage-free calibration, and posterior-uncertainty experiments remain open (sec. 11.19).

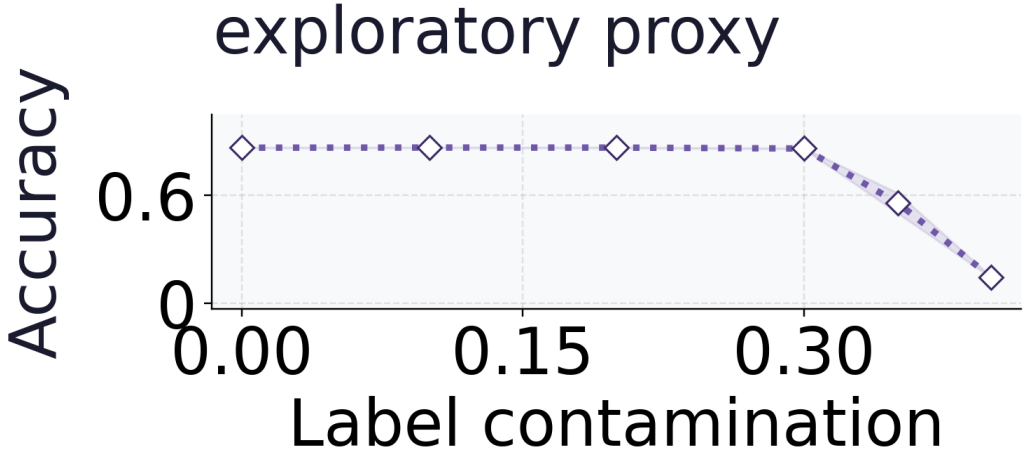
Exploratory... (part 16)



Exploratory... (part 17)

nll / L2=0.05 (standard proxy)

Exploratory… (part 18)



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Exploratory... (part 19)

rcce / L2=0.10 (exploratory proxy)

Exploratory... (part 20)

Composite NLL/L2=0.05 versus RCCE/L2=0.10 point-estimate proxy. The joint change cannot identify an RCCE-only effect.

Exploratory... (part 21)

peak_margin_contamination is selected within the displayed contamination sweep by maximum composite-configuration held-out

Exploratory... (part 22)

accuracy margin; robust_loss_param and n_per are configured inputs, not selected by this report

Exploratory... (part 23)

Largest displayed margin: 2.8% at 35% contamination.

Exploratory… (part 24)

Shaded bands retain the source-supplied intervals. They do not represent FedGVI posterior uncertainty.

Exploratory... (part 25)

Exploratory proxy only: no Alpha-Renyi objective, calibration, universal robustness, or exact source-protocol replication.

Exploratory... (part 26)

fig. 21 is exploratory per-client proxy evidence. Its loss-limit check, finite synthetic contrast, and the separate source-conditional theorem have different roles; none comes from the server comparison of sec. 10.5.2.

Exploratory... (part 27)

The authoritative three-axis boundary is sec. 6.3.

Exploratory... (part 28)

PyTorch deterministic-MLP complement (executed). The optional pipeline also instantiates generalized variational inference with a point-mass MLP family: Linear$\to$ReLU$\to$Linear$\to$softmax with 16 hidden units. Clients use the density-power β-loss at $\beta = 0.5$ for 200 Adam steps,

Exploratory... (part 29)

and per-test-point predictions are fused with `robust_aggregate` at `robustness = 0.5` under PyTorch 2.12.1.

Exploratory... (part 30)

The executed consensus is a valid probability simplex, with maximum unit-mass deviation 2.22e-16, and repeated seeded runs are bit-identical (Yes). At contamination 0.40, held-out consensus accuracy is 0.558 for the $\beta \to 0$ client and 0.545 for the $\beta = 0.5$ client.

Exploratory... (part 31)

This single-seed endpoint demonstrates API transfer and deterministic execution, not posterior-uncertainty inference, model-class universality, or a robust-loss advantage at neural-network scale. It is separate from the 64-seed exploratory logistic-regression curve above and from the source-conditional theorem.

Exploratory... (part 32)

When PyTorch is absent, the pipeline records unavailable-value sentinels; builds exercising this optional lane install the `torch` extra (sec. 13).